

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall generate weekly meal plans strictly adhering to user macronutrient constraints and allergen exclusion lists.	High	Pass: Generated menu contains zero excluded allergen ingredients. Fail: Peanuts included for peanut-allergic profile.	Core functionality to ensure patient safety and adherence to basic dietary needs.
FR-002	Functional	The system shall recalculate the subscriber's next week's macro targets based on logged adherence and weight-trend data.	High	Pass: Next week's macros adjust automatically based on logs. Fail: Macros remain static despite logged changes.	Ensures the diet plan remains effective and adapts to progress.
FR-003	Functional	The system shall allow an assigned Dietitian to review, edit, or override an auto-generated plan before dispatch.	Medium	Pass: Dietitian edits are saved and reflected in the user's plan. Fail: Auto-generated plan dispatches ignoring edits.	Provides professional oversight for automated plans.
FR-004	Functional	The system shall allow a Subscriber to pause, skip, or reschedule deliveries up to 12 hours before the dispatch cutoff window.	Medium	Pass: User skips a delivery, and remaining totals adjust to meet targets. Fail: Targets are missed without automatic adjustment.	Offers flexibility while maintaining dietary goals.
FR-005	Functional	The system shall propose a valid substitute ingredient when a planned item is unavailable, subject to confirmation.	Low	Pass: System suggests substitute meeting macros/allergens. Fail: Substitute violates constraints.	Ensures service continuity during supply issues.

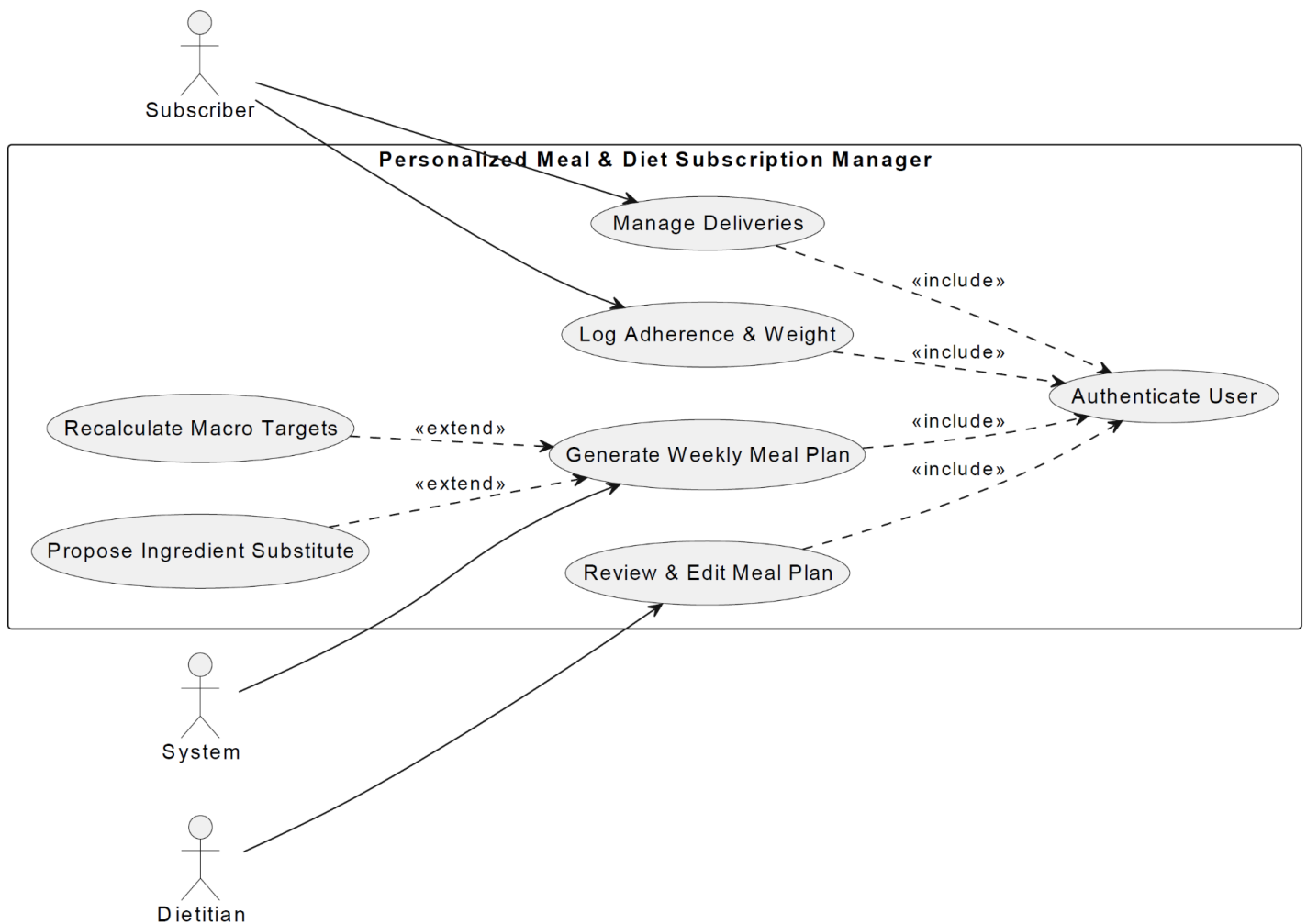
NFR-001	Nonfunctional	The system shall process schedule modifications (pause, skip) in under 5 seconds during peak server load.	High	Pass: Benchmarking confirms < 5s latency for updates. Fail: Latency exceeds 5s causing UI timeouts.	Ensures timely updates for logistics and dispatch.
NFR-002	Nonfunctional	The system shall encrypt all patient metabolic data at rest and restrict access to the subscriber and assigned dietitian.	High	Pass: Audit confirms no unauthorized read access to records. Fail: Unauthorized roles can query raw data.	Complies with health data privacy regulations.

Actors:

- **Subscriber:** The patient receiving the customized meal plans.
- **Dietitian:** The medical professional reviewing and approving the plans.
- **System:** The automated subscription and calculation engine.

Use Cases:

- **UC-01:** Authenticate User
- **UC-02:** Generate Weekly Meal Plan
- **UC-03:** Manage Deliveries (pausing, skipping, or rescheduling)
- **UC-04:** Log Adherence & Weight
- **UC-05:** Review & Edit Meal Plan
- **UC-06:** Recalculate Macro Targets (Extends UC-02)
- **UC-07:** Propose Ingredient Substitute (Extends UC-02)



Use Case: Generate Weekly Meal Plan

Preconditions:

- The subscriber has an active subscription and a completed health profile (allergies, diabetic limits, macro goals).
- The system is authenticated to access the subscriber's private health data.
- Inventory data for ingredients is up to date.

Main Success Scenario:

1. The scheduled time for weekly meal plan generation is reached for a given subscriber.
2. The system securely retrieves the subscriber's current macronutrient targets and allergen exclusion lists.
3. The system fetches logged adherence and weight-trend data from the previous week.
4. The system recalculates the next week's macro targets based on the fetched logs (Includes <> UC-06: Recalculate Macro Targets).
5. The system selects recipes and ingredients that strictly fulfill the recalculated macros without violating allergen exclusions.
6. The system compiles the selections into a 7-day meal plan.
7. The system saves the generated meal plan as "Pending Review" in the database.
8. The system sends a notification to the assigned Dietitian that a new plan is ready for review.

Alternate Flow: Ingredient Unavailable

- In Step 5: A planned ingredient for a selected recipe is out of stock in the inventory system.
 1. The system identifies the unavailable ingredient.
 2. The system identifies a substitute ingredient that also satisfies the subscriber's allergen exclusions and macro targets (Includes <> UC-07: Propose Ingredient Substitute).
 3. The system flags the substitution in the meal plan, requiring Dietitian confirmation.
 4. The system resumes from Step 6 of the Main Success Scenario.